

E-COMMERCE SALES ANALYTICS DASHBOARD

Project Documentation

Power BI | DAX | Data Visualization

1. Project Overview

This project is an interactive E-Commerce Sales Analytics Dashboard built in Power BI. I designed it as a three-page report so that the most important business information can be understood quickly, while still allowing a user to explore customers and products in more detail. The dashboard brings sales, profit, orders, customers, products, categories, regions and segments together in one place.

2. What I Wanted to Achieve

The main goal of the dashboard was to turn the sales data into something that is easy to read and useful for decision-making. In particular, I wanted the report to help answer questions such as:

- How are overall sales and profit performing?
- Which customers and products are contributing the most?
- Which regions and customer segments generate more sales?
- Which categories and sub-categories are performing well?
- How much business comes from repeat customers?
- Which customers have strong sales as well as healthy profit margins?

3. Tools I Used

Tool / Technology	How I used it
Power BI Desktop	Built the dashboard, visuals and report pages.
DAX	Created the KPIs and analytical measures used in the dashboard.
Power BI Visuals	Used KPI cards, line charts, donut charts, bar charts and tables to present the results.
Page Navigator	Made it easy to move between the Executive, Customer and Product pages.

4. How the Dashboard Is Organized

Page 1 — Executive Dashboard

This page gives a quick overall picture of the business. It shows Total Sales, Total Quantity, Total Profit, Total Orders and Profit Margin, followed by monthly sales trends, sales by segment, Top 10 products, regional sales and category sales.

Page 2 — Customer Analytics

This page looks at the customer side of the business. It includes Total Customers, Repeat Customers, Average Orders per Customer and Average Sales per Customer. It also shows customer segment performance, the Top 10 customers by sales and profit, and a profitability table.

Page 3 — Product Analytics

This page focuses on products and categories. It shows Total Products, Total Categories, Total Sub-Categories and Average Sales per Product, along with sales by category and region, the Top 10 sub-categories and the Top 10 products.

5. Main KPIs

Page	Value	KPI
Executive	1.57M	Total Sales
Executive	22K	Total Quantity
Executive	175.26K	Total Profit
Executive	3,003	Total Orders
Executive	11.19%	Profit Margin
Customer	773	Total Customers
Customer	689	Repeat Customers
Customer	3.88	Average Orders per Customer
Customer	2.03K	Average Sales per Customer
Product	1,755	Total Products
Product	3	Total Categories
Product	17	Total Sub-Categories
Product	892.20	Average Sales per Product

6. Page 1 — Executive Dashboard

The Executive Dashboard is designed to be the first page a manager would look at. It provides the big-picture numbers first and then adds a few visuals to explain where those numbers come from.

- The monthly sales line chart makes it easy to see how sales change throughout the year.
- The Sales by Segment donut chart shows the contribution of each customer segment.
- The Top 10 Products chart highlights the products with the highest sales.
- The Sales by Region chart compares West, East, Central and South.
- The Sales by Category chart compares Office Supplies, Technology and Furniture.

7. Page 2 — Customer Analytics

The Customer Analytics page focuses on who is driving the business and how profitable those customers are.

- There are 773 total customers, of which 689 are repeat customers.
- The average number of orders per customer is 3.88, while average sales per customer is 2.03K.
- Consumer is the largest customer segment, contributing approximately 753K (48.09%) in sales.
- Corporate contributes approximately 509.74K (32.55%), and Home Office contributes approximately 303.06K (19.35%).
- The Top 10 customer charts make it easy to compare customers by sales and by profit.
- The profitability table brings customer name, total sales, total profit and customer profit margin together for easier comparison.

8. Page 3 — Product Analytics

The Product Analytics page helps explain which parts of the product portfolio are contributing most to sales.

- The dashboard contains 1,755 products across 3 categories and 17 sub-categories.
- Office Supplies is the highest-sales category in the displayed category comparison.
- Phones are the leading sub-category in the Top 10 Sub-Categories by Sales chart.
- The Top 10 Products by Sales chart shows the strongest individual products.
- The regional sales chart provides an additional geographic view of performance.

9. What the Dashboard Shows

A few important observations stand out from the completed dashboard:

- Total sales are approximately 1.57M, while total profit is approximately 175.26K.
- The overall profit margin shown on the Executive Dashboard is 11.19%.
- Consumer customers account for the largest share of sales at 48.09%.
- West is the strongest region in the displayed regional comparison, while South is the lowest.
- Office Supplies is the highest-sales category.
- Phones are the leading sub-category in the Top 10 Sub-Categories chart.
- Caroline Jumper appears as the top customer by sales.
- Tamara Chand appears as the top customer by profit.
- 3D Systems Cube Printer appears as the leading product in the Top 10 Products chart.

10. Business Recommendations

Based on the patterns visible in the dashboard, the analysis can be used to support the following actions:

- Focus on retaining repeat customers and strengthening relationships with high-value customer segments.
- Look closely at the strongest products and sub-categories when planning inventory and promotions.
- Investigate lower-performing regions and categories and consider targeted sales activities.
- Use customer profit margin together with sales instead of judging customer value from revenue alone.
- Use the monthly sales trend to help plan future campaigns and allocate resources.

11. Interactivity and Navigation

To make the report easier to use, I added a Page Navigator connecting all three pages: Executive Dashboard, Customer Analytics and Product Analytics. Users can move between the pages without having to manually search through report tabs. The charts and tables can also be explored interactively to understand customer, product, category, segment and regional performance.

12. Conclusion

Overall, this project helped turn a large sales dataset into a dashboard that is easier to understand and use. The three pages work together: the Executive Dashboard gives the overall picture, Customer

Analytics explains customer value and profitability, and Product Analytics explains product and category performance. The result is a single interactive report that can be used to monitor performance and identify areas that may need attention.

13. Short Project Description for Resume or Portfolio

Built a three-page interactive E-Commerce Sales Analytics Dashboard using Power BI and DAX.

Analyzed sales, profit, orders, customers, products, categories, sub-categories, segments and regions through KPI cards, trend analysis, Top 10 rankings, customer profitability analysis and interactive page navigation.